

Informe de experimento: pfagent

Generado el 2026-09-16 13:59 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio pfagent (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f  
↪ low-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent
```

- Reports leídos: 1 (report.rescored.json usado en 1; el resto report.json)
- Modelos: openrouter:openai/gpt-5.4
- Métodos: pfagent
- Casos: case14
- Requests: generadas, N=40 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=10, parameterized=10, multistep=10, ambiguous=10
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-16 13:36
- Costo total: 1.6125 USD (40 items; costo medio 0.04031 USD/item)
- Items totales: 40; ok=40; con error=0; tokens totales=567798
- Solved global: 62.5 % (25/40)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

Método pfagent

System prompt (SYSTEM_PROMPT_EN) —1375 caracteres; architecture=react, gate=False, memory=False

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step
```

User prompt (primer item) = request_text —67 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

Load case14 and disconnect the line between bus ten and bus eleven.

3. Resultados agregados

Columnas del protocolo (medias por método $\times$ caso, del `scoreboard_per_case`), en los 4 grupos de `metricas_pfactagent_definicion`, **utilidad de tarea** Form. (formulación exacta de las tool calls), V_MAE (p.u.) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (números de la respuesta trazables a salidas de tools) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Abst. (abstención en items resolubles) y \$ (costo total USD). Porcentajes en %.

Por método $\times$ caso

Method	Case	N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Abst.	V pass	\$
pfagent	case14	40	62.5 (25/40)	3.17e-04	0.06	62.5 (25/40)	100.0 (40/40)	8.26e-03	100.0	–	2.8	14195–	2.5 (1/40)	–	100.0 (40/40)	1.6125

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	plain	parameterized	multistep	ambiguous
pfagent	80.0 (8/10)	90.0 (9/10)	40.0 (4/10)	40.0 (4/10)

Solved por dificultad (%, items resueltos / items)

Method	plain	parameterized	multistep	ambiguous
pfagent	80.0 (8/10)	90.0 (9/10)	40.0 (4/10)	40.0 (4/10)

Verificación final (pfagent: $V(x,c,z,y)$ aplicado a la respuesta final)

`pass_first` = pasó al primer intento; `pass_retry` = pasó tras el único reintento con el veredicto añadido; `abstained` = falló dos veces, respuesta forzada a fallo declarado sin números; `retry_mutation` = el reintento intentó una herramienta que muta la red (bloqueada, nunca ejecutada; cuenta como abstención) —ver “Integridad del reintento” abajo. N = items con `final_gate` activo (no todos los métodos lo tienen).

Method	N	pass_first	pass_retry	abstained	retry_mutation
pfagent	40	90.0 (36/40)	7.5 (3/40)	2.5 (1/40)	0.0 (0/40)

Integridad del reintento

De 4 ítem(s) que llegaron a un reintento, 0 intentaron una herramienta que muta la red (bloqueada) en vez de corregir el reporte; se cuentan como abstención, no como “pasó”. El mensaje de reintento ya no sugiere la corrección específica (p. ej. “reconnect...”) y bloquea estructuralmente `modify_load`, `disconnect_line`, `reconnect_line`, `apply_remedial_action` y `load_case` mientras el reintento está en efecto.

Tipos de error de formulación (conteo de items)

Method	extra_arg	changes_result	missed_step	ok	wrong_id
pfagent	13		1	25	1

Errores de ejecución por método (conteo de items con error)

(ningún item con *error*)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (`claimed_success_on_failure`, `stale_state`, `abstained`). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de `traces/` cuando existe.

Método pfagent

Ejemplo: resuelto (solved=True)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": 5930 chars más}
gate: PASS (reasons=, mismatch=7.85e-09 MW)
--- Ronda 3 ---
LLM: 1. One-sentence summary
I loaded IEEE case14 and disconnected the line between bus 10 and bus 11; the power flow converged successfully.

2. Key numbers
- Total load: 253.85 MW
- Total generation: 266.51 MW
- Total losses: 12.66 MW
- Voltage range: 1.010 pu to 1.090 pu

3. Violations, if a ...[538 chars más]
--- Respuesta final ---
1. One-sentence summary
I loaded IEEE case14 and disconnected the line between bus 10 and bus 11; the power flow converged successfully.

2. Key numbers
- Total load: 253.85 MW
- Total generation: 266.51 MW
- Total losses: 12.66 MW
- Voltage range: 1.010 pu to 1.090 pu

3. Violations, if any
- Voltage violations:
  - Bus 1: 1.060 pu, overvoltage
  - Bus 6: 1.070 pu, overvoltage
  - Bus 7: 1.057 pu, overvoltage
  - Bus 8: 1.090 pu, overvoltage
  - Bus 11: 1.064 pu, overvoltage
  - Bus 12: 1.054 pu, overvoltage
- Thermal violations: none
- The disconnected line 10-11 now has 0.0 MW flow, confirming it is out of service.

4. Suggested next step
If you want, I can now check which branch is most heavily loaded after this outage, run an N-1 contingency analysis, or recommend remedi ...
↳ [38 chars más]
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=5347;
↳ cost_usd=0.0173; wall=5.4 s
```

Ejemplo: formulación fallida (formulation_exact=False)

```
request (case14-ambiguous-011-s0, case14, difficulty=ambiguous, seed=1971893210): Load case14 and set the active load at bus 6 to 14,700 kw.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 6, "p_mw": 14.7})
executed_calls: load_case({"case_name": "case14"}) -> get_status({}) -> modify_load({"bus_id": 6, "p_mw": 14.7, "q_mvar": 11.2})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 263.13580543670497,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: get_status({})
output: {"active_case": "case14", "network_info": {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11,
↳ "total_load_mw": 263.13580543670497, "total_gen_capacity_mw": 772.4000000000001}, "has_last_result": false}
gate: n/a (no es un resultado de flujo)
--- Ronda 3 ---
tool: modify_load({"bus_id": 6, "p_mw": 14.7, "q_mvar": 11.2})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.236323374534588, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 6297 chars más}
gate: PASS (reasons=, mismatch=6.47e-09 MW)
--- Ronda 4 ---
LLM: 1. I loaded IEEE case14 and set the active load at bus 6 to 14,700 kW, which solved successfully.

2. Key numbers:
- Total load: 267.145 MW
- Total generation: 281.819 MW
```

```

3. Violations, if any:
- Voltage violations only; no thermal vio ...[248 chars más]
--- Respuesta final ---
1. I loaded IEEE case14 and set the active load at bus 6 to 14,700 kW, which solved successfully.

2. Key numbers:
- Total load: 267.145 MW
- Total generation: 281.819 MW
- Total losses: 14.674 MW
- Voltage range: 1.010 pu to 1.090 pu

```

```
3. Violations, if any:
- Voltage violations only; no thermal violations were reported.
- Overvoltage buses: 1, 6, 7, 8, 9, 10, 11, 12

4. Suggested next step:
- If you want, I can now check the most loaded branch, run an N-1 contingency analysis, or suggest remedial actions to reduce the
  ↳ overvoltage violations.
--- Scoring ---
formulation=extra_arg_changes_result (exact=False); V_MAE=9.52e-17; F_MAE=0.00; faithful=100.0 %; solved=False (formulation); flags=-;
↳ error=-; tokens=6531; cost_usd=0.0194; wall=5.4 s
```

```
request (case14-multistep-030-s0, case14, difficulty=multistep, seed=537958268): Load case14, set the active load at bus 14 to 17.2 MW,  
→ disconnect the line between bus 1 and bus 5, reconnect the line between bus 1 and bus 5, and finally run an N-1 contingency analysis  
→ ranked by min voltage and report the 8 worst outages.  
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 14, "p_mw": 17.2}) -> disconnect_line({"from_bus": 1,  
→ "to_bus": 5}) -> reconnect_line({"from_bus": 1, "to_bus": 5}) -> run_n1_contingency({"top_k": 8, "criteria": "min_voltage"})  
executed_calls: load_case({"case_name": "case14"}) -> run_powerflow({}) -> modify_load({"bus_id": 14, "p_mw": 17.2, "q_mvar": 0}) ->  
→ disconnect_line({"from_bus": 1, "to_bus": 5}) -> reconnect_line({"from_bus": 1, "to_bus": 5}) -> run_n1_contingency({"top_k": 8,  
→ "criteria": "min_voltage", "max_candidates": 0}) -> run_powerflow({})  
fuente de la traza: archivo de traza completo  
--- Ronda 1 ---  
tool: load_case({"case_name": "case14"})  
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 271.9322468585065,  
→ "total_gen_capacity_mw": 772.4000000000001}  
gate: n/a (no es un resultado de flujo)  
--- Ronda 2 ---  
tool: run_powerflow({})  
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,  
→ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.373522845971019, "is_violation": false,  
→ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va ...[6356 chars más]  
gate: PASS (reasons=-, mismatch=5.81e-09 MW)  
--- Ronda 3 ---  
tool: modify_load({"bus_id": 14, "p_mw": 17.2, "q_mvar": 0})  
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,  
→ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.4514286734574515, "is_violation": false,  
→ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "v ...[6408 chars más]  
gate: PASS (reasons=-, mismatch=5.92e-09 MW)  
--- Ronda 4 ---  
tool: disconnect_line({"from_bus": 1, "to_bus": 5})  
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,  
→ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -8.382663773008685, "is_violation": false,  
→ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0099999 ...[6332 chars más]  
gate: PASS (reasons=-, mismatch=2.66e-08 MW)  
--- Ronda 5 ---  
tool: reconnect_line({"from_bus": 1, "to_bus": 5})  
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,  
→ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.4514286734574515, "is_violation": false,  
→ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "v ...[6408 chars más]  
gate: PASS (reasons=-, mismatch=5.92e-09 MW)  
--- Ronda 6 ---  
tool: run_n1_contingency({"top_k": 8, "criteria": "min_voltage", "max_candidates": 0})  
output: {"plot_type": "n1_ranking", "figure_json":  
→ "{\n\"data\": [{\n\"claponaxis\": false,\n\"hoverinfo\": \"text\",\n\"hovertext\": [\n\\\"\\u003cb\\\"\\u003eOutage\\\"\\u003c\\\"\\u002fb\\\"\\u003e: 1-5\n→ (\nline\\\"\\u003cbr\\\"\\u003e\\\"\\u003cb\\\"\\u003eConverged\\\"\\u003c\\\"\\u002fb\\\"\\u003e: True\\\"\\u003cbr\\\"\\u003e\\\"\\u003cb\\\"\\u003eVoltage\n→ violations\\\"\\u003c\\\"\\u0 ...[19527 chars más]\n"}  
gate: n/a (no es un resultado de flujo)  
--- Ronda 7 ---  
LLM: 1. **Summary:** I loaded IEEE case14, changed bus 14 active load to 17.2 MW, disconnected and then reconnected line 1-5, and finally  
→ ran an N-1 contingency analysis on the restored network ranked by minimum voltage for the 8 worst outages.
```

LLM: 1. ****Summary:**** I verified the final restored case14 state after setting bus 14 active load to 17.2 MW and reconnecting line 1-5, and
 ↳ the N-1 analysis ranked by minimum voltage identifies outage ****1-5**** as the worst of the 8 requested outages.

```

2. **Key numbers:**
  - Total load: **275.1414299554 ...[1722 chars más]
--- Respuesta final ---
No numerical result is reported. Verification failed: the answer quotes 2 number(s) from a solve made before the last network change
  ↳ (reconnecting the branch between bus 1 and bus 5). This is a declared failure of the verification step, not a claim that the network
  ↳ failed to converge.
--- Scoring ---
formulation=extra_arg_changes_result (exact=False); V_MAE=1.62e-03; F_MAE=0.08; faithful=-- %; solved=False (formulation);
  ↳ flags=abstained,abstained_on_solvable; error=-; tokens=99412; cost_usd=0.2674; wall=25.6 s

```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- $V_MAE \geq 1e-4$ p.u. en 1 filas método×caso con solver: **pfagent**/case14 (3.17e-04).
- El gate verificó 58 resultados de flujo, 0 fallaron la verificación y se activó (retuvo números) 0 veces en 0 items.
- Ningún item terminó con **error**.
- Solved de **pfagent**: 62.5 % (25/40).
- Formulación exacta agregada sobre métodos con tools: 62.5 % (25/40).
- Formulación por dificultad: mejor **parameterized** 90.0 % (9/10), peor **multistep** 40.0 % (4/10).
- Errores de formulación más frecuentes: **extra_arg_changes_result=13**, **missed_step=1**, **wrong_id=1**.
- Ningún item reclamó éxito sobre un fallo (**claimed_success_on_failure=0**).
- 0 de 25 items con mutación de red reportaron estado obsoleto (**stale_state**).
- Trazabilidad de números (Faith.) media con tools: 100.0 %.
- Costo total 1.6125 USD; el método más caro fue **pfagent** con 1.6125 USD (100 %).
- Métodos con tools: 2.83 tool calls medias por item (máximo 7).

6. Archivos

Rutas absolutas bajo **pfagent**. **report.rescored.json** (cuando existe) es el archivo leído para las métricas.

```

report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
  ↳ /power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent/report.json [leído:
  ↳ report.rescored.json]
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent/scoreboard_per_case.json
scoreboard.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent/scoreboard.rescored.json
scoreboard_per_case.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent/scoreboard_per_case.rescored.json
traces (40 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent/traces

```